Quantum Fault Tolerance Meets Toom's Contours

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1 Toric Encoded Circuits

Definition 1.1 (Troic Encoded Circuit)**.** Let C be a quantum circuit. We say that C is a *Toric encoded circuit* if the following holds:

1. There is a Toric code $\mathcal{C}$ such that the qubits of $\mathcal{C}$ can be partitioned into registers $R_1, \ldots, R_k$, and, at any time, each of the registers is encoded either by the *nice encoding of* $\mathcal{C}$ or by the *nice encoding of* $\phi \circ H(\mathcal{C})$[1]. At time $t = 0$, the registers are initialized to be either $|\bar{0}\rangle$ or encoded $|\mathbf{S}\rangle$.

2. The layers of C at even times are sweep-cycles, and the layers at odd times are realizations of one of the logical $\mathbf{H}, \mathbf{S}, \mathbf{CX}, \mathbf{X}$, or $\mathbf{Z}$ gates.

Definition 1.2 (The Boundery of Deviation ∂D_t)**.** Let C be a Toric encoded circuit, with registers $R_1, .., R_l$ and let $\mathcal{E}$ be a set of faults. Consider the subjection $C[\mathcal{E}]$, For a time t, let $A_t, B_t \subset \{R_1, .., R_l\}$ be the registers that encoded, at time t in $C[\mathcal{E}]$, by $\mathcal{C}$ and $\phi \circ \mathbf{H}$ respectively. Denote by $\mathcal{D}_t$ the deviation in $C[\mathcal{E}]$ at time t. We define the *X-type, and the Z-type boundaries* of $\mathcal{D}_t$, denoted by ∂D_t^X and ∂D_t^Z, as follow:

$$\partial \mathcal{D}_t^X := \bigcup_{R_i \in A_t} \partial \left| R_{i,t} \right\rangle \cup \bigcup_{R_i \in B_t} \partial \left(\phi \circ \mathbf{H} \right) \left| R_{i,t} \right\rangle$$
$$\partial \mathcal{D}_t^Z := \bigcup_{R_i \in B_t} \partial \left| R_{i,t} \right\rangle \cup \bigcup_{R_i \in A_t} \partial \left(\phi \circ \mathbf{H} \right) \left| R_{i,t} \right\rangle$$

Definition 1.3 (Base graph G_o)**.** Let C be a quantum circuit, at depth d, with registers $R_1, \ldots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, .., \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C, denoted by $G_o = (V_o, E_o)$ as follows:

[1] Notice that although the code space is invariant to the application of $\phi \circ \mathbf{H}$, the positive direction of the underlying Toric complex is flipped.

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \dots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:

$$V_o := \bigcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \dots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t+1$ that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$\begin{aligned} E_o := \Big\{ &\{(v,t),(u,t')\} : \text{ if} \\ &\text{either there exists } i \text{ such } \{v,u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\ &\text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1}+1 \Big\} \end{aligned}$$

[COMMENT] Here we should add that a vertex v belongs to location l if there is a qubit f that v belongs to its associated face.

Definition 1.4 (Tooms Contour Graph of $C(\mathcal{E})$)**.** For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o. For each time t, denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$. We define the *Tooms contour graph* $\mathcal{H} = (V_{\mathcal{H}}, E_{\mathcal{H}})$ as follows:

1. The vertices $V_{\mathcal{H}}$ are partitioned into two types, X-type vertices $V_{\mathcal{H},X}$ and Z-type vertices $V_{\mathcal{H},Z}$. Each of the X/Z-type corresponds to the vertices supporting the X/Z-boundary of $\mathcal{D}$. Namely:

$$\begin{aligned} V_{\mathcal{H},X} &:= \left\{ v_t : v_t = (v,t) \in V_o \text{ and there exists a face } f \in \partial\mathcal{D}_t^X \text{ such } v \in f \right\} \\ V_{\mathcal{H},Z} &:= \left\{ v_t : v_t = (v,t) \in V_o \text{ and there exists a face } f \in \partial\mathcal{D}_t^Z \text{ such } v \in f \right\} \end{aligned}$$

Similarly, the Z-type vertices $V_{\mathcal{H},Z}$ are the vertices $(v,t) \in V_o$, that satisfy that v belongs to a face associated with a check, which, at time t, belongs to the $(\partial D_t)\,|_Z$.

2. The edges $E_{\mathcal{H}}$ is partitioned into two types $A_{\mathcal{H}}$ and $F_{\mathcal{H}}$, where for any logical gate g which applied in time t in $C[\mathcal{E}]$ we add edge to $A_{\mathcal{H}}$ as follow:

   (a) If g is a logical **CX** then for any physical **CX** in its realization, between faces f_1 into f_2 at time t we add:

$$\begin{aligned} &\left\{ \{(u,t),(v,t+1)\} : u \in f_1, v \in f_1 \cup f_2 \text{ and } v,u \in V_{\mathcal{H},X} \right\} \\ &\left\{ \{(u,t),(v,t+1)\} : u \in f_2, v \in f_1 \cup f_2 \text{ and } v,u \in V_{\mathcal{H},Z} \right\} \end{aligned}$$

   (b) If g is logical $\phi \circ \mathbf{H}$, then for any physical $\mathbf{H}$, in its realization, that acts on face f, we add:

$$\begin{aligned} &\left\{ \{(u,t),(v,t+1)\} : u \in f, v \in \phi(f) \text{ and } v \in V_{\mathcal{H},Z}, u \in V_{\mathcal{H},X} \right\} \\ &\left\{ \{(u,t),(v,t+1)\} : u \in f, v \in \phi(f) \text{ and } v \in V_{\mathcal{H},X}, u \in V_{\mathcal{H},Z} \right\} \end{aligned}$$

(c) If g is one of logical X, Z, I or a sweep-cycle, then for any face f, in the register on which acts g, we add:

$$\big\{\{(u,t),(v,t+1)\} : u,v \in f \text{ and } u,v \in V_{\mathcal{H},X}\big\}$$
$$\big\{\{(u,t),(v,t+1)\} : u,v \in f \text{ and } u,v \in V_{\mathcal{H},Z}\big\}$$

In addition for any fault g of $\mathcal{E}$, that is applied at time t, and any faces $f_1 \notin \mathrm{Sup}_X(g)$ and $f_2 \notin \mathrm{Sup}_Z(g)$, we add the edges:

$$\big\{\{(u,t),(v,t+1)\} : u,v \in f_1 \text{ and } u,v \in V_{\mathcal{H},X}\big\}$$
$$\big\{\{(u,t),(v,t+1)\} : u,v \in f_2 \text{ and } u,v \in V_{\mathcal{H},Z}\big\}$$

To construct $F_{\mathcal{H}}$, we connect any pair of vertices (v,t), (u,t) if there exists a vertex $(w,t+1)$ such both of them are connected to it trough $A_{\mathcal{H}}$. Namely:

$$F_{\mathcal{H}} = \big\{e = \{u,v\} \in E_o : \text{ exists } w \in V_{\mathcal{H}} \text{ such } \{v,w\},\{u,w\} \in A_{\mathcal{H}}\big\}$$

We call $A_{\mathcal{H}}$ and $F_{\mathcal{H}}$ the *arrows* and *forks* edges respectively. By definition of the construction, $G_{\mathcal{H}}$ is a time-graph.